EE 354/CE 361/MATH 310 -

Introduction to Probability and Statistics

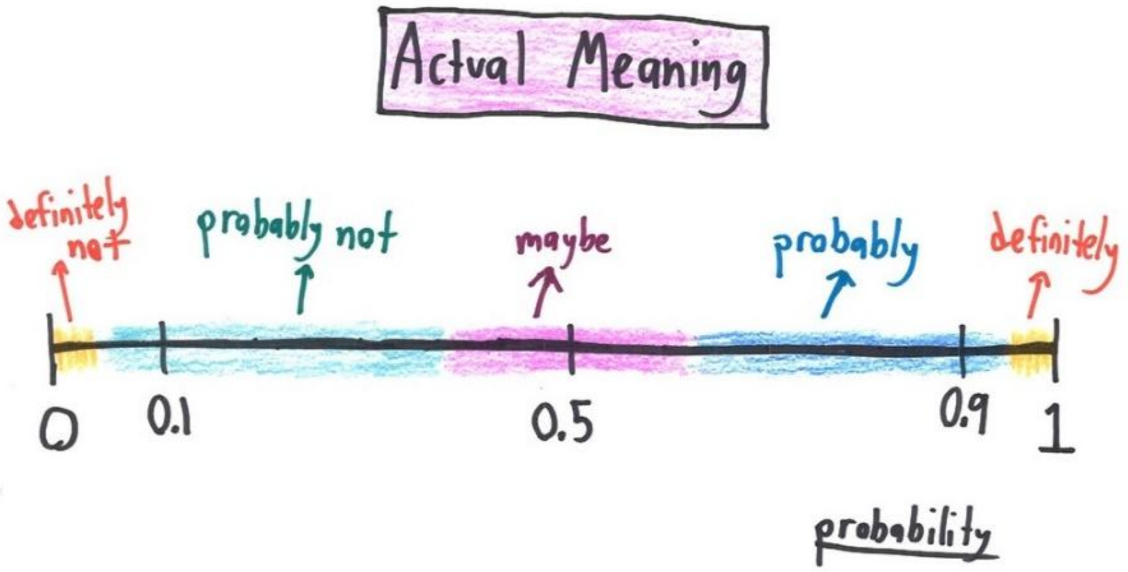
Spring 2026

EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

Aamir Hasan

Week 2 - Lecture No 03 – January 19, 2026



Announcements!

1. **Quiz No 01 – January 21, 2026**
2. **Office hours – Tuesday & Thursday 12:00 to 1:00 PM @ Room C-211 (or through meeting request)**
3. **Supplemental Reading on Set Theory uploaded**

Agenda for today

1. Unit 1 – Probability models and axioms (will try to complete this today)

Unit 1: Probability models and axioms – Lecture outline

- Sample space
- Probability laws
  - Axioms
  - Some properties
- Examples
  - Discrete
  - Continuous
- Interpretations of probabilities

Probability calculation steps (sequence of 4 steps) - Recap

1. Specify the sample space

2. Specify the probability law

3. Identify an event of interest

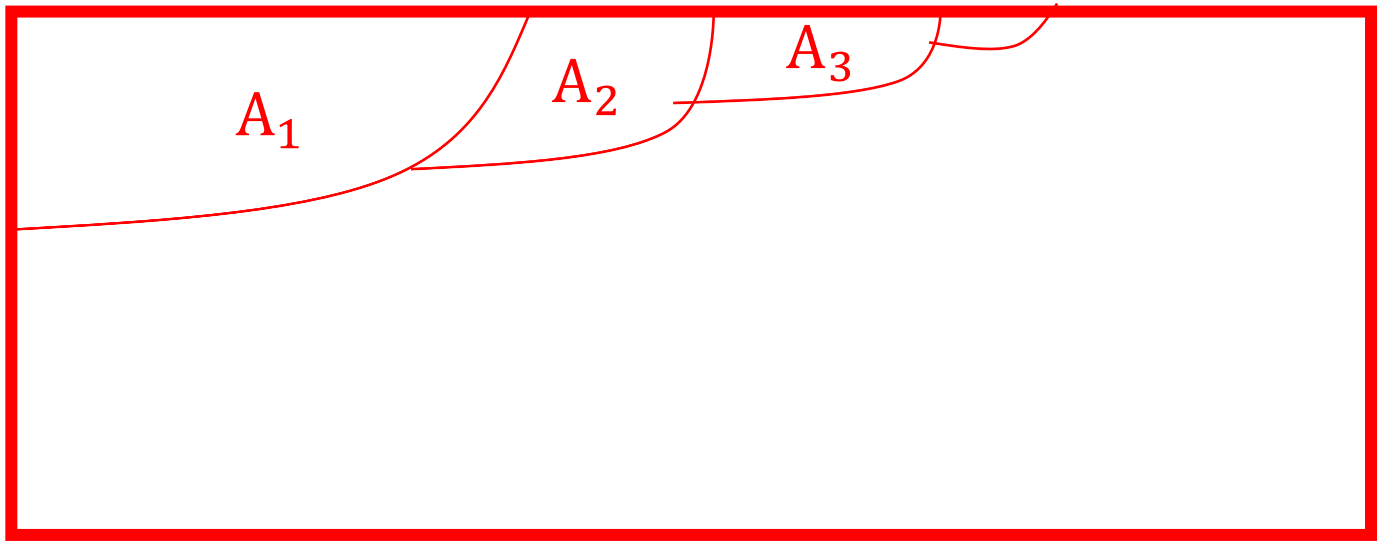
4. Calculate…

Countable additivity axiom

- Strengthens the finite additivity axiom

Countable Additivity Axiom:

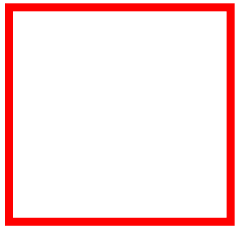
If $A_1, A_2, A_3, \ldots$ is an infinite sequence of disjoint events,
then $P\,(A_1 \cup A_2 \cup A_3 \cup \ldots) = P(A_1) + P(A_2) + P(A_3) + \ldots.$



Countable additivity axiom

Countable Additivity Axiom:

If $A_1, A_2, A_3, \ldots$ is an infinite sequence of disjoint events,
then $P\,(A_1 \cup A_2 \cup\ A_3 \cup\ \ldots) = P(A_1) + P(A_2) + P(A_3) + \ \ldots.$

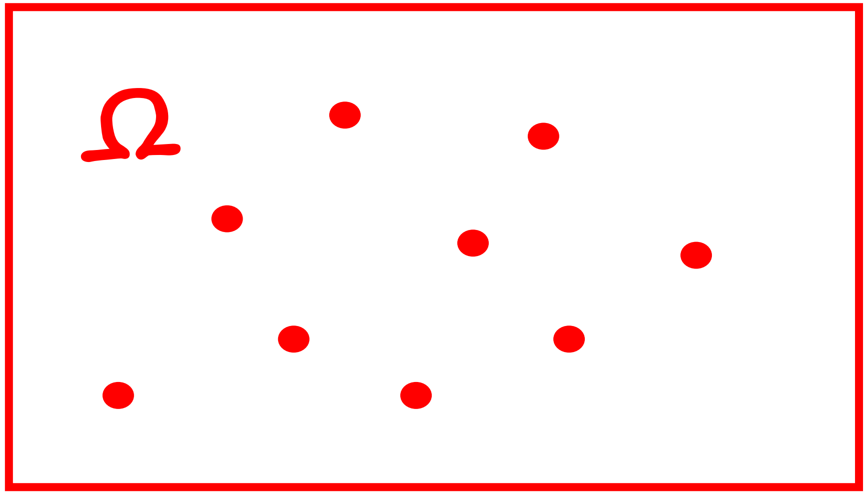


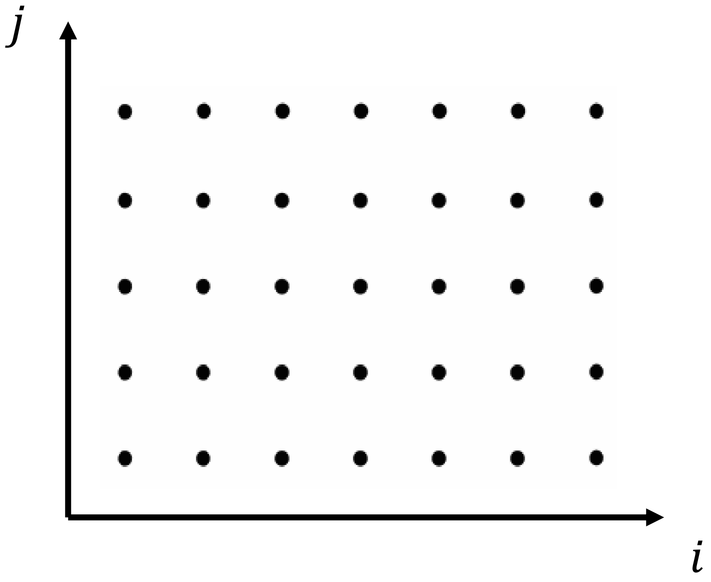
$$1 = P\,(\Omega) = P\,(\cup\,\{(x,y)\}) \stackrel{?}{=} \sum P\,(\{(x,y)\}) = \sum 0 = 0$$

- Additivity holds only for **"countable"** sequences of events
- The unit square (similarly, the real line, etc.) is **not countable** (its elements cannot be arranged in a sequence)
- "Area" is a legitimate probability law on the unit square, as long as we do not try to assign probabilities/areas to "very strange" sets

Countable versus uncountable sets

- Countable – can be put in a 1-1 correspondence with positive integers
  - Positive integers?
  - All integers?
  - Pair of integers?
- Uncountable –
  - Unit interval [0, 1]



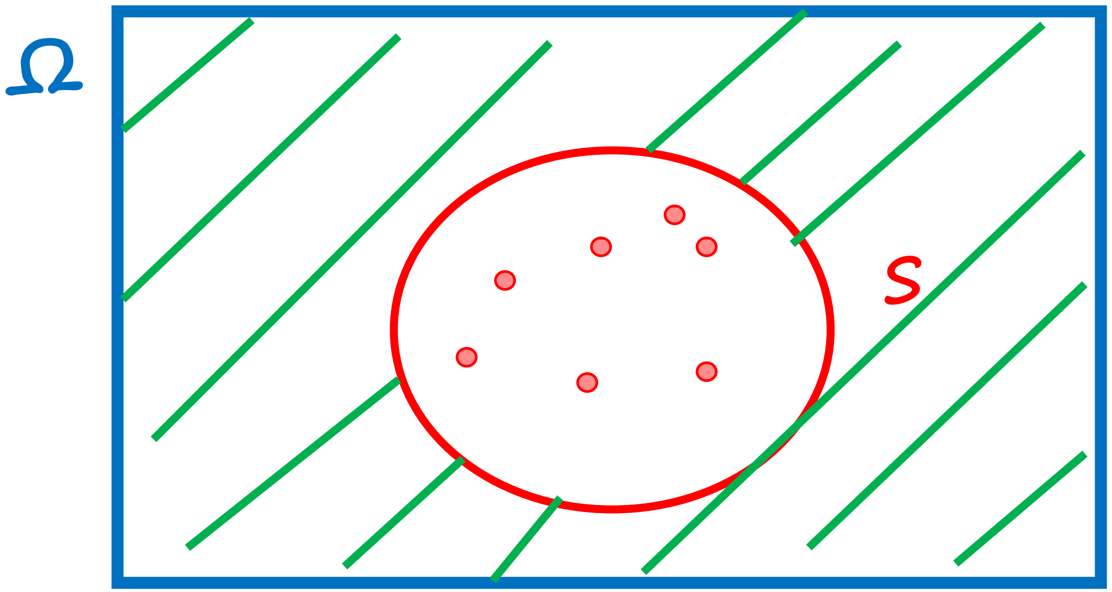


Unit 1: Mathematical Background (Self-Study)

- Sets & De Morgan’s law
- Infinite Geometric series
- Sums with multiple indices
- Countable & uncountable sets
- Bonferroni inequality

Sets

- A collection of distinct elements



$\{a, b, c, d\}$ finite

R: real numbers infinite

$\{ x \in \mathrm{R} : \cos(x) > 1/2 \}$

$x \in S$ $x \notin S$

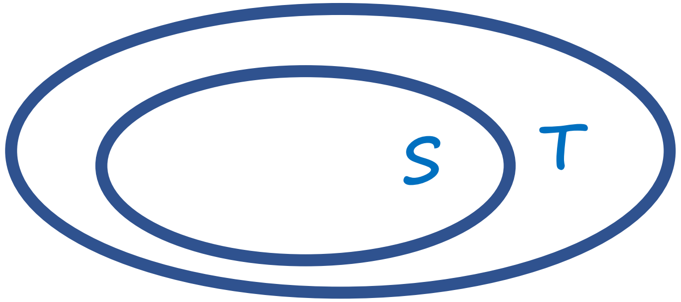
Ω = universal set

S^c $x \in S^c$ if $x \in \Omega$, $x \notin S$

$\emptyset$ = empty set

$\Omega^c = \emptyset$

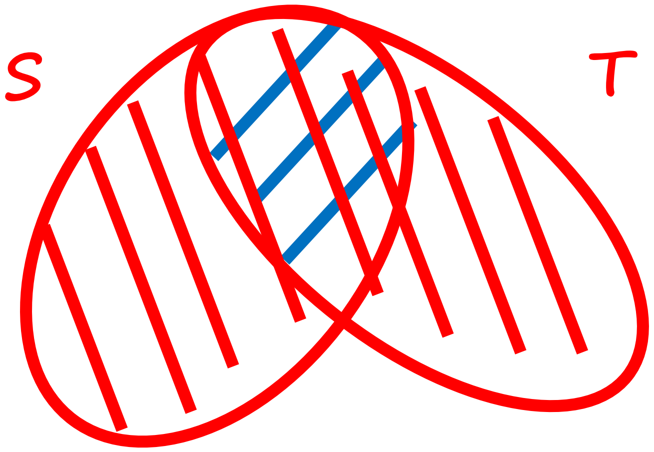
$(S^c)^c = S$



$S \subset T : x \in S \Rightarrow x \in T$

$\subseteq$

Unions and intersections

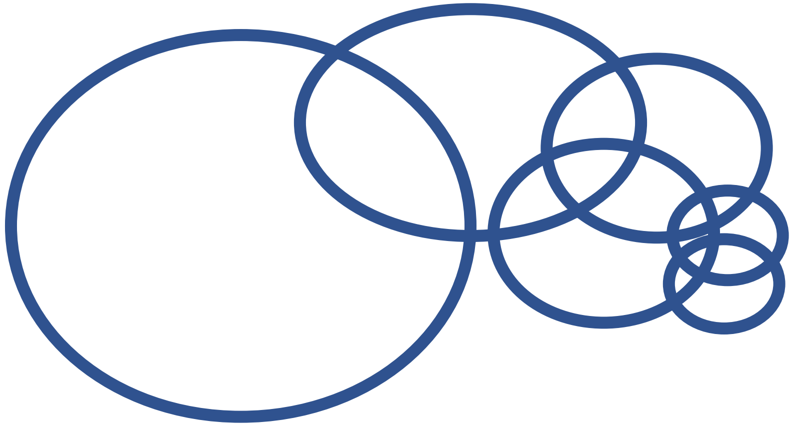


$S \cup T$

$S \cap T$

$x \in S \cup T \Leftrightarrow x \in S$ or $x \in T$

$x \in S \cap T \Leftrightarrow x \in S$ and $x \in T$



S_n $\quad$ $n=1,2,\ldots$

$x \in \bigcup_n S_n$ iff $x \in S_n$, for some n

$x \in \bigcup_n S_n$ iff $x \in S_n$, for all n

Set properties

$S \cup T \cup U$

$\shortparallel$

- $S \cup T = T \cup S$
- $S \cap (T \cup U) = (S \cap T) \cup (S \cap U)$
- $(S^c)^c = S,$

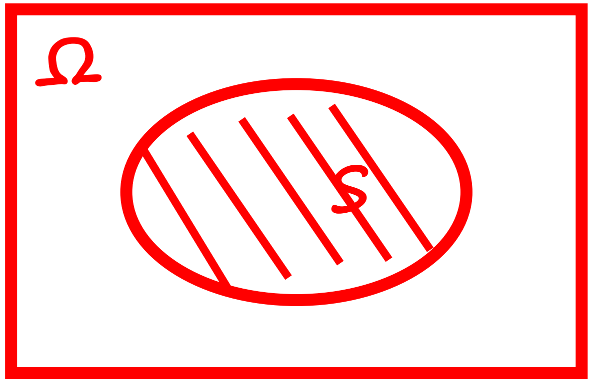
$S \cup \Omega = \Omega,$

$S \cup (T \cup U) = (S \cup T) \cup U$

- $S \cup (T \cap U) = (S \cup T) \cap (S \cup U)$

$S \cap S^c = \emptyset,$

$S \cap \Omega = S.$



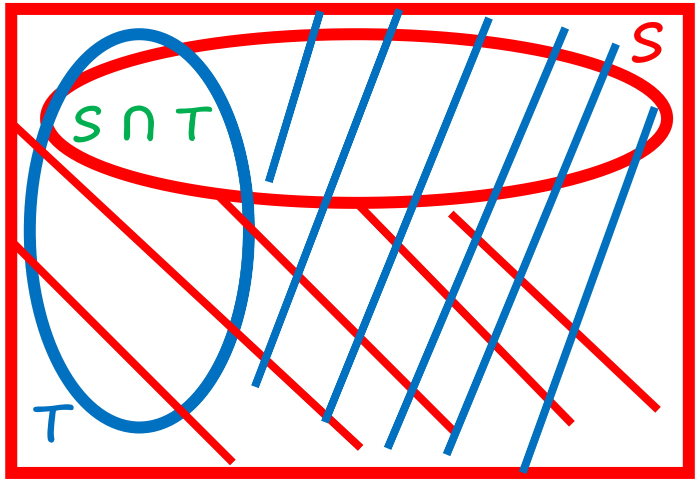
$S \cap (T \cap U) = (S \cap T) \cap U$

$\left.\begin{array}{l} S \subset T \\ T \subset S \end{array}\right\} \Rightarrow S = T$

De Morgan's laws

$S \cap (T \cap U) = (S \cap T) \cap U$

$$\boxed{(S \cap T)^c = S^c \cup T^c}$$



$S \to S^c \qquad T \to T^c$

$S^c \to S \qquad T^c \to T$

$(S^c \cap T^c)^c = S \cup T$

$$\boxed{S^c \cap T^c = (S \cup T)^c}$$

$$\left(\bigcap_n S_n\right)^c = \bigcup_n S_n{}^c$$

$$\left(\bigcup_n S_n\right)^c = \bigcap_n S_n{}^c$$

$$x \in (S \cap T)^c \Leftrightarrow x \notin S \cap T \Leftrightarrow \left\{\begin{array}{c} x \notin S \\ \text{or} \\ x \notin T \end{array}\right\} \Leftrightarrow \left\{\begin{array}{c} x \in S^c \\ \text{or} \\ x \in T^c \end{array}\right\} \Leftrightarrow x \in S^c \cup T^c$$

Example - Geniuses + Chocolates

$P\,(G) = 0.6$ $\quad$ $P\,(C) = 0.7$ $\quad$ $P\,(G \cap C) = 0.4$

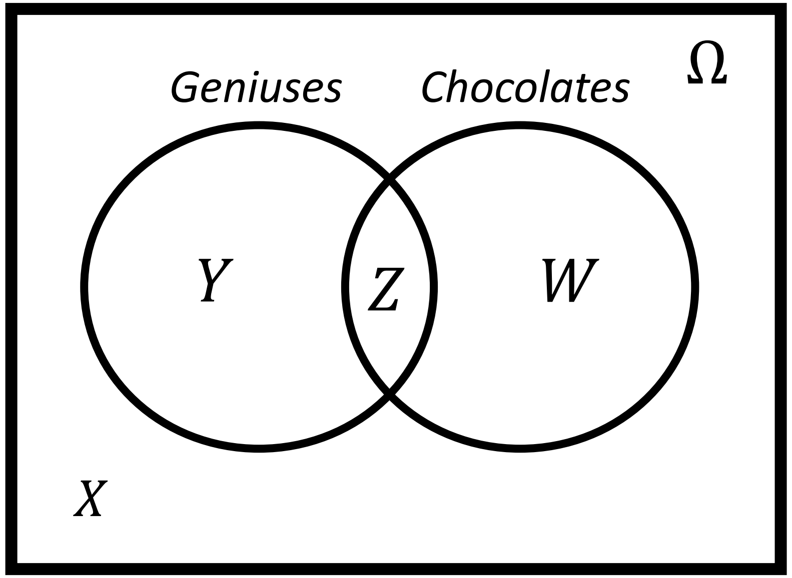
Probability that student picked up at random neither likes chocolate nor is a genius?

$$P\,(G) = P\,(Y \cup Z) = P\,(Y) + P\,(Z) = 0.6 => P\,(Y) = 0.2$$

$$P\,(C) = P\,(Z \cup W) = P\,(Z) + P\,(W) = 0.7 => P\,(W) = 0.3$$

$$P\,(Z) = 0.4$$

Class - EE 354



$$1 = P\,(\Omega) = P\,(X) + P\,(Y) + P\,(Z) + P\,(W)$$

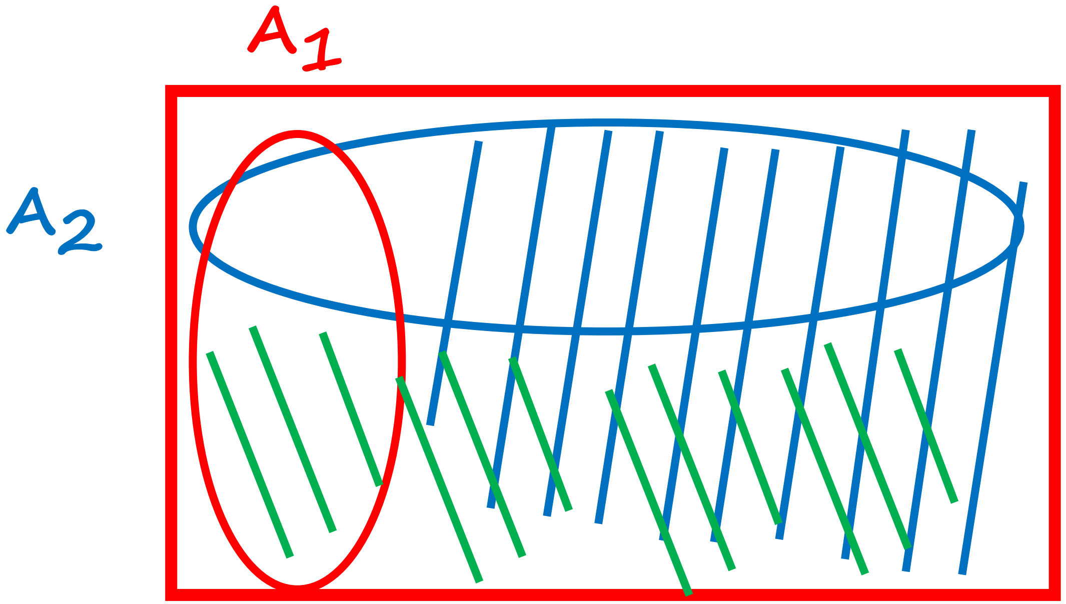
$$= P\,(X) + 0.2 + 0.4 + 0.3$$

$$=> P\,(X) = 0.1$$

Interpreting the union bound and the Bonferroni inequality

$$P\,(A_1 \cup A_2) \leq P(A_1) + P(A_2)$$

- Suppose that:
  - Very few students are smart A_1
  - Very few students are beautiful A_2
- Then: very few students are smart or beautiful



- Suppose that:
  - Most of the students are smart
  - Most students are beautiful
- Then: most students are smart and beautiful

$$P\,(A_1 \cap A_2) \geq \underline{P(A_1)} + \underline{P(A_2)} - 1$$

The Bonferroni inequality

$$\boldsymbol{P}\,(\boldsymbol{A_1} \cap \boldsymbol{A_2}) \geq \boldsymbol{P}(\boldsymbol{A_1}) + \boldsymbol{P}(\boldsymbol{A_2}) - \mathbf{1}$$

$$\boldsymbol{P}\,(\boldsymbol{A_1} \cup \boldsymbol{A_2}) \leq \boldsymbol{P}(\boldsymbol{A_1}) + \boldsymbol{P}(\boldsymbol{A_2})$$

$$P\,(A_1 \cap A_2)^c = P\left(A_1^c \cup A_2^c\right) \leq P\left(A_1^c\right) + P\left(A_2^c\right)$$

$$\cancel{1} - P\,(A_1 \cap A_2) \qquad \leq \cancel{1} - P(A_1) + 1 - P(A_2)$$

$$\underline{\boldsymbol{P}\,(\boldsymbol{A_1} \cap \ldots \cap \boldsymbol{A_n})} \geq \underline{\boldsymbol{P}(\boldsymbol{A_1}) + \cdots + \boldsymbol{P}(\boldsymbol{A_n})} - \underline{\underline{(\boldsymbol{n} - \mathbf{1})}}$$

$$P\,(A_1 \cap \ldots \cap A_n)^c = P\left(A_1^c \cup \ldots \cup A_n^c\right) \leq P\left(A_1^c\right) + \ldots + P(A_n^c)$$

$$1 - P\,(A_1 \cap \ldots \cap A_n) \quad \leq \quad (1 - P(A_1)) + \ldots + (1 - P(A_n))$$

Interpretations of probability theory

- A narrow view: a branch of math
  - ➢ Axioms → theorems

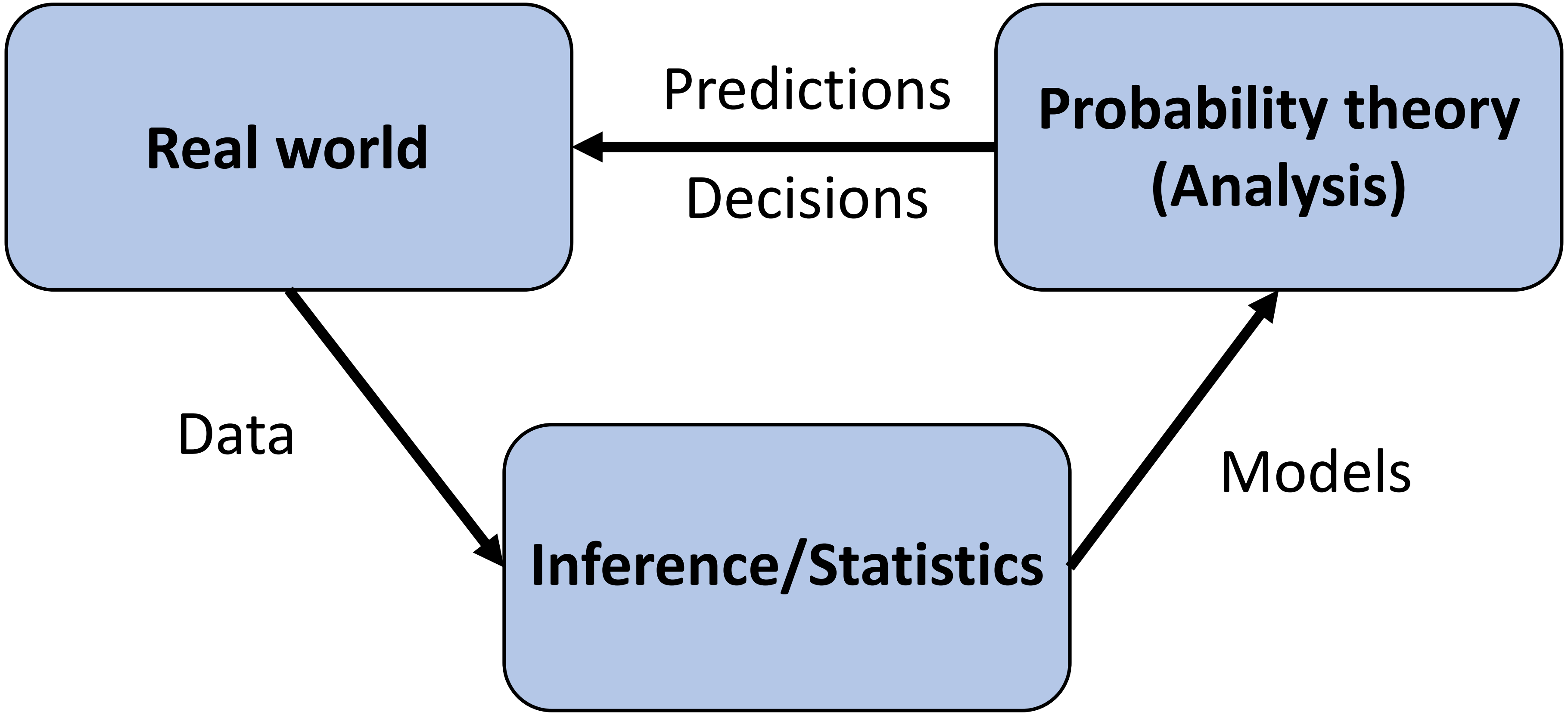
"Thm:" "Frequency" of event A "is" P (A)

- Are the probabilities frequencies?
  - ➢ **P**(coin toss yields heads) = ½
  - ➢ **P**(Mr. Imran Khan will be reelected as the PM in the next general elections in Pakistan) = 0.7

- Probabilities are often interpreted as:
  - ➢ Description of beliefs
  - ➢ Betting preferences

Interpretations of probability theory

- A framework for analyzing phenomena with uncertain outcomes
  - Rules for consistent reasoning
  - Used predictions and decisions



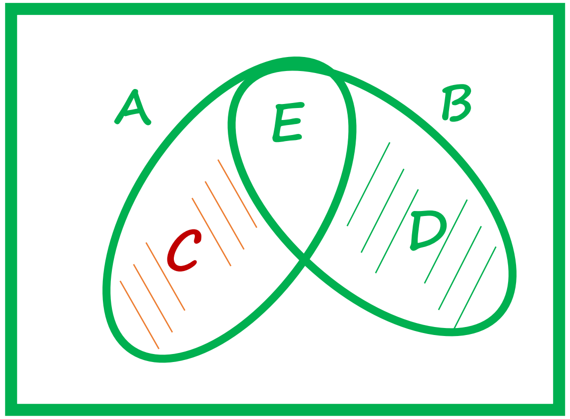
4 Practise Problems

Probability calculation steps (sequence of 4 steps)

1. Specify the sample space
2. Specify the probability law
3. Identify an event of interest
4. Calculate…

Practice Problem 1

Show $P(C \cup D) = P(A) + P(B) - 2P(E)$



$$P(C \cup D) = P(C) + P(D)$$

$$= P(C) + P(E) + P(D) + P(E) - 2P(E)$$

$$= P(A) + P(B) - 2P(E)$$

Practice Problem 2

Tim has a peculiar four-sided dice. When he rolls the dice, the probability of getting any particular outcome is proportional to result of the dice.

(a) What is the probability of the number being even?

Sample Space

Probability Assignment

Practice Problem 2

Tim has a peculiar four-sided dice. When he rolls the dice, the probability of getting any particular outcome is proportional to result of the dice.

(a) What is the probability of the number being even?

Sample Space *Probability Assignment*

No	Probability
1	p
2	$2p$
3	$3p$
4	$4p$

$10p = 1$

$p = 1/10$

Sample Space Probability Assignment

No	Probability
1	$1/10$
2	$2/10$
3	$3/10$
4	$4/10$

P(even No) = 6/10

Practice Problem 3 – *do it in pairs*

Hasan has a peculiar pair of four-sided dice. When he rolls the dice, the probability of getting any particular outcome is proportional to the sum of the results of the two individual dice.

(a) What is the probability of the sum being even?

(b) What is the probability of Hasan rolling a 2 and a 3, in any order?

Practice Problem 3 - Solution

Hasan has a peculiar pair of four-sided dice. When he rolls the dice, the probability of getting any particular outcome is proportional to the sum of the results of the two individual dice.

(a) What is the probability of the sum being even?

(b) What is the probability of Hasan rolling a 2 and a 3, in any order?

Die 1	Die 2	Sum	$\mathbf{P}$(Sum)
1	1	2	2p
1	2	3	3p
1	3	4	4p
1	4	5	5p
2	1	3	3p
2	2	4	4p
2	3	5	5p
2	4	6	6p
3	1	4	4p
3	2	5	5p
3	3	6	6p
3	4	7	7p
4	1	5	5p
4	2	6	6p
4	3	7	7p
4	4	8	8p
		Total	80p

$$\mathbf{P}(\text{All outcomes}) = 80p \text{ (Total from the table)}$$

and therefore

$$p = \frac{1}{80}$$

(a)

$$\mathbf{P}(\text{Even sum}) = 2p + 4p + 4p + 6p + 4p + 6p + 6p + 8p = 40p = \boxed{1/2}$$

(b)

$$\mathbf{P}(\text{Rolling a 2 and a 3}) = \mathbf{P}(2,3) + \mathbf{P}(3,2) = 5p + 5p = 10p = \boxed{1/8}$$

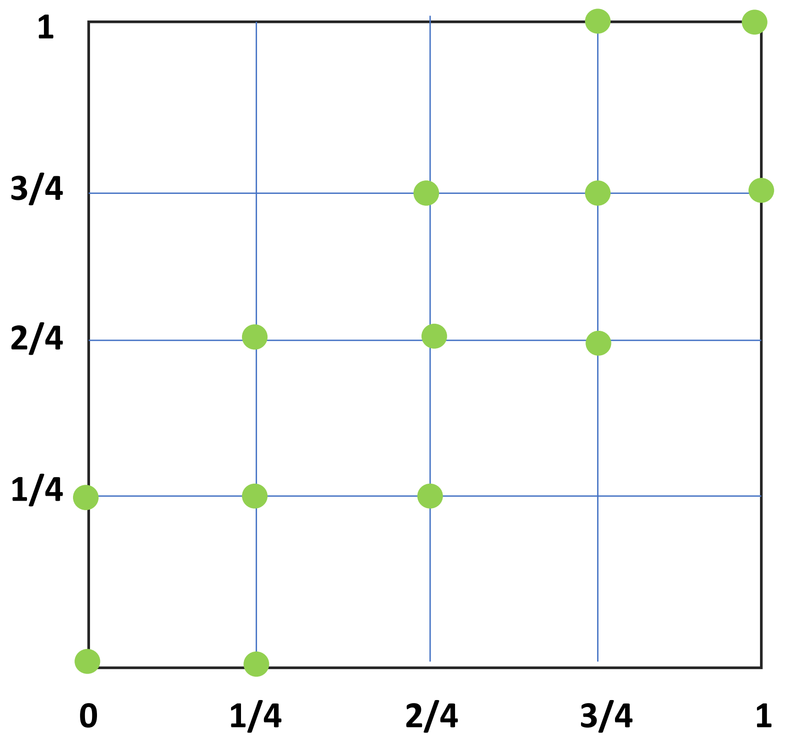
Practice Problem 4

"S" and "A" have a meeting at a given time and each will arrive at the meeting place with a delay between 0 and 1 hour – **all pairs of delay being equally likely**. The first to arrive will wait for 15 minutes and will leave if the other has not yet arrived. **What is the probability that they will meet?**

Two Cases of equally likely times –

A) Arrival times are Discrete (equally likely) for both 'S' & 'A' - in steps of 15 mins (t = 0, 1/4 , 2/4, 3/4 , 1)

B) Arrival times are Continuous



P(Meeting) = 13/25

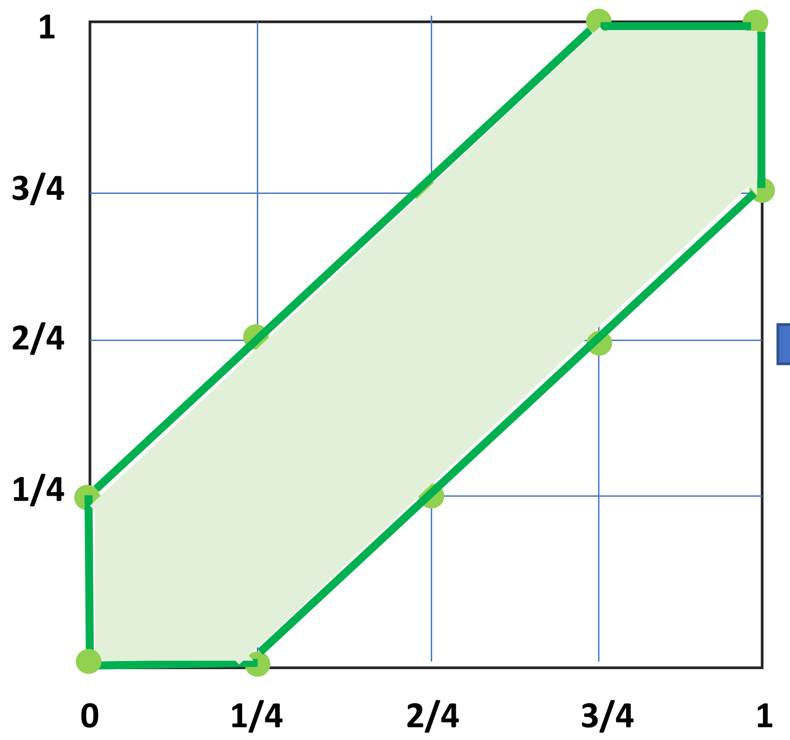
Practice Problem 4

"S" and "A" have a meeting at a given time and each will arrive at the meeting place with a delay between 0 and 1 hour – **all pairs of delay being equally likely**. The first to arrive will wait for 15 minutes and will leave if the other has not yet arrived. **What is the probability that they will meet?**

Two Cases of equally likely times –

A) Arrival times are Discrete (equally likely) for both 'S' & 'A' - in steps of 15 mins (t = 0, 1/4 , 2/4, 3/4 , 1)

B) Arrival times are Continuous



$$\mathrm{P(Meeting)} = 1 - [2 \times (1/2 \times 3/4 \times 3/4)]$$
$$= 1 - 9/16$$
$$= 7/16$$

Conditioning & Bayes' rule

- Let's say that you roll two dice and their sum comes out to be S.
  - ❖ If S = 9, what can we say about the first and second roll?
    - ✓ What is the probability that first roll is 1 or 2?

New knowledge should be used to modify our original beliefs.

↓

Revised probabilities are called the conditional probabilities.

Some mathematical concepts (Self-study)

Mathematical background: Geometric Series

$$S = \sum_{i=0}^{\infty} \propto^i = 1 + \propto + \propto^2 + \ldots = \frac{1}{1-\propto} \qquad |\propto| < 1$$

$$(1 - \propto)(1 + \propto + \ldots + \propto^n) = 1 - \propto^{n+1}$$

$n \to \infty$

$$(1 - \propto) S = 1$$

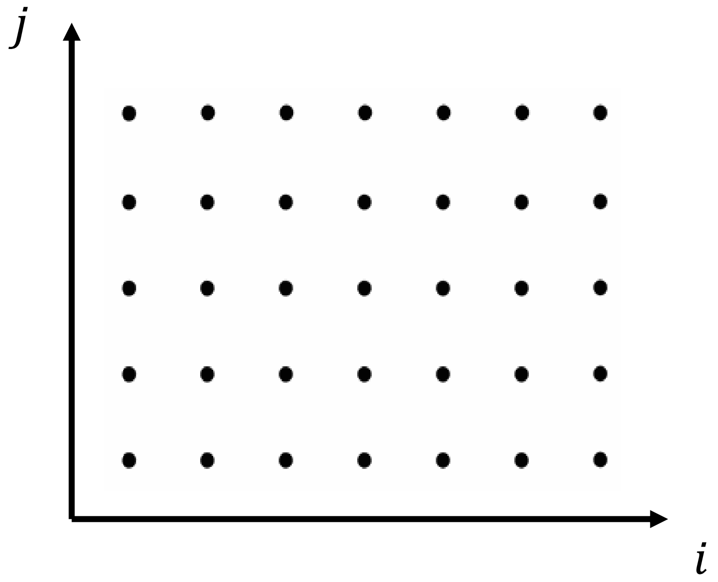
$$S = 1 + \sum_{i=1}^{\infty} \propto^i = 1 + \propto \sum_{i=0}^{\infty} \propto^i = 1 + \propto S \Rightarrow S(1 - \propto) = 1$$

$S < \infty$ *taken for granted*

About the order of summation in series with multiple indices

$$\sum_{i\geq 1, j\geq 1} a_{ij}$$

$n \rightarrow \infty$



(a) Fix i and let j => 1 - ∞

$$\sum_{i=1}^{\infty}\left(\sum_{j=1}^{\infty} a_{ij}\right)$$

(b) Fix j and let i => 1 - ∞

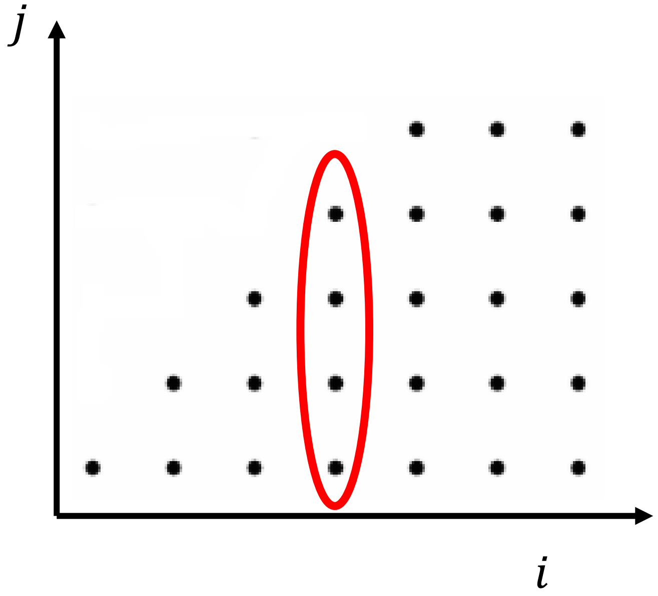
$$\sum_{j=1}^{\infty}\left(\sum_{i=1}^{\infty} a_{ij}\right)$$

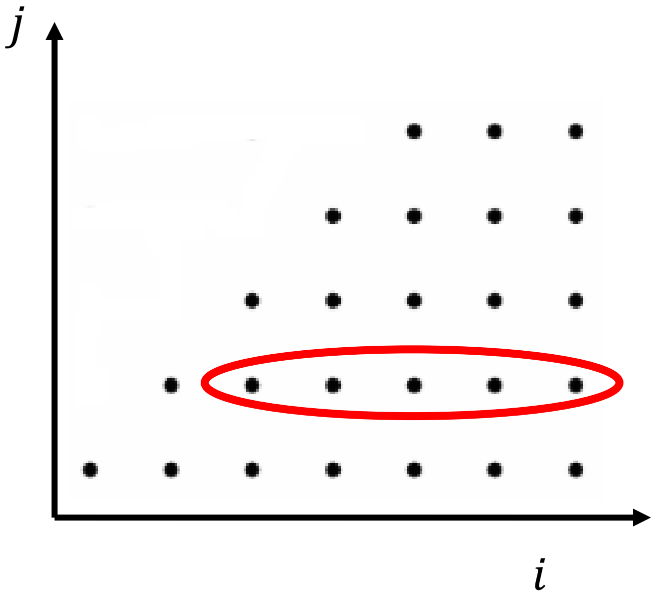
if $\sum |a_{ij}| < \infty$, then order does not matter

About the order of summation in series with multiple indices

if

$$\sum_{(i,j):j\leq i} |a_{ij}| < \infty$$





$$\sum_{i=1}^{\infty}\sum_{j=1}^{i} a_{ij} = \sum_{j=1}^{\infty}\sum_{i=j}^{\infty} a_{ij}$$